

Adaptive Shared Control for Collaborative Robots via Vision-Based Motion Prediction and Ergonomic Evaluation

Abstract—As human-robot collaboration (HRC) becomes increasingly prevalent in dynamic environments, seamlessly blending human intent with autonomous robotic execution remains a critical challenge. Standard motion prediction models often exhibit overconfidence or fail to account for human ergonomic comfort, leading to jerky robotic responses and operator fatigue. This paper proposes a novel adaptive shared control framework designed for collaborative robot arms. By integrating a deep learning-based motion predictor with an adaptive weight generator, the system explicitly fuses the reliability of algorithmic predictions with real-time human arm comfort scores. A trajectory smoother dynamically allocates control authority, ensuring stable and fluent cooperation. The proposed control loop, implemented via a quaternion-based inverse-kinematics formulation and an optimal LQR controller, strictly constrains absolute joint positions to ensure robust tracking. Experimental validation on a physical collaborative robot demonstrates significant improvements in task fluency, reduced control effort, and enhanced human postural comfort compared to traditional fixed-authority baselines.

Index Terms—Human-Robot Collaboration, Shared Control, Deep Learning, Motion Prediction, Quaternions.

I. INTRODUCTION

THE integration of collaborative robots (cobots) into shared workspaces requires advanced control paradigms that can seamlessly adapt to human behavior. While vision-based skeleton tracking and deep learning (DL) models have significantly improved human motion prediction, relying solely on autonomous predictions can be dangerous if the algorithm's confidence drops due to occlusion or sudden movements.

Existing literature often overlooks the necessity of dynamically blending robot autonomy with human ergonomic requirements. To address this gap, this paper introduces an adaptive shared control architecture. The key contributions of this work are threefold:

- 1) A novel adaptive weight generation mechanism that fuses DL prediction reliability with real-time arm comfort scores mapped against an optimal human pose dataset.
- 2) A robust trajectory smoothing filter that ensures stable control authority blending without inducing oscillatory behavior in the cobot.
- 3) A rigorously constrained, quaternion-based control formulation validated on a physical robot platform, demonstrating statistical superiority over fixed-weight blending methods.

II. RELATED WORK

A. Human Motion Prediction in HRC

Recent advancements in vision-based systems have enabled real-time skeleton tracking. However, standard DL approaches often lack confidence bounds, making them susceptible to anomalous predictions during dynamic tasks.

B. Authority Blending and Shared Control

Traditional shared control systems typically utilize static authority weights or rely on simplistic distance-based heuristics. Dynamic weight allocation that incorporates human physiological states remains under-explored.

III. SYSTEM ARCHITECTURE OVERVIEW

The proposed control framework processes inputs from an AI-enabled vision system to extract the human wrist position $\mathbf{p}_{wr}$ and human joint angles θ_{hu} . A DL-based motion prediction module estimates the future trajectory $\hat{\mathbf{p}}_{pre}$ and its associated error ϵ .

Simultaneously, the Adaptive Weight Generator evaluates algorithmic reliability and arm comfort, synthesizing a blending weight w . The Trajectory Smoother then fuses the predicted trajectory with the current robot feedback $\mathbf{p}_{fb}$ to generate a smooth command $\mathbf{p}_{sm}$, which is subsequently tracked by the robot via an optimal LQR controller.

IV. PROPOSED METHODOLOGY

A. Vision-based Motion Prediction

The motion prediction module utilizes a recurrent neural network architecture to process sequential skeleton data. Let $\mathbf{p}_{wr}(t)$ be the actual wrist position captured by the camera at time t , and $\hat{\mathbf{p}}_{pre}(t)$ be the model's prediction.

B. Adaptive Weight Generator

The core novelty of the framework lies in the dynamic generation of the blending weight w , derived from two distinct metrics.

First, the prediction reliability is quantified by the instantaneous prediction error:

$$\epsilon(t) = \|\hat{\mathbf{p}}_{pre}(t) - \mathbf{p}_{wr}(t)\|_2 \quad (1)$$

The corresponding reliability score $s_r(t)$ is defined using an exponential decay function:

$$s_r(t) = \exp(-\lambda\epsilon(t)), \quad \lambda > 0 \quad (2)$$

where λ governs the sensitivity to prediction errors.

Second, the arm comfort score s_j is computed by comparing the current human joint configuration θ_{hu} against an optimal pose dataset θ_{opt} to minimize biomechanical stress.

The final dynamic weight w is a linear fusion of these two scores:

$$w = \alpha \cdot s_r + \beta \cdot s_j \quad (3)$$

where α and β are tunable hyperparameters reflecting the relative importance of algorithmic confidence and human ergonomics.

C. Trajectory Smoothing and Shared Control

To prevent discontinuous commands from reaching the robot controller, authority blending is formulated as a discrete-time smoothing filter for a future horizon step k :

$$\mathbf{p}_{sm}(t+k) = w \cdot \hat{\mathbf{p}}_{pre}(t+k) + (1-w) \cdot \mathbf{p}_{fb} \quad (4)$$

This formulation guarantees asymptotic stability during the transition of control authority.

D. Robust Robot Control and Kinematics

To resolve the inverse kinematics (IK) from the smoothed trajectory $\mathbf{p}_{sm}$ to the reference joint configuration q_{ref} , this framework employs a pseudo-inverse Jacobian method. Crucially, end-effector orientation is represented strictly using unit quaternions rather than Euler angles. This eliminates representation singularities and ensures smooth rotational tracking during complex manipulative tasks.

Furthermore, to guarantee robustness in the numerical integration step, the IK algorithm is designed to directly constrain the final resulting joint angle vector q against mechanical limits, rather than solely bounding the differential change Δq . The resulting reference q_{ref} is then tracked by an LQR controller to compute optimal joint velocities $\dot{q}_{cur}$.

V. EXPERIMENTAL SETUP

The proposed architecture was validated using a high-fidelity setup featuring an AI-capable camera for skeleton tracking and a physical collaborative robot arm. The software backend was developed using cross-platform C/C++ build systems managed via CMake to ensure real-time execution.

VI. RESULTS AND DISCUSSION

To evaluate the effectiveness of the proposed framework, ablation studies were conducted against two baselines:

- 1) Fixed weight blending ($w = \text{const}$).
- 2) Reliability-only blending (ignoring comfort score s_j).

Results indicate that the proposed adaptive weight generator significantly reduces the root-mean-square error (RMSE) of the trajectory tracking while minimizing sharp torque gradients at the robot joints. Furthermore, analysis of the operator's posture demonstrated a measurable improvement in overall arm comfort over extended operational periods.

VII. CONCLUSION

This paper presented an adaptive shared control framework for human-robot collaboration that uniquely fuses deep learning prediction reliability with human ergonomic comfort. By ensuring mathematically stable authority blending and enforcing strict quaternion-based kinematic constraints, the proposed system yields safer, more fluent collaborative performance. Future work will investigate the integration of haptic feedback and the exploration of multi-agent collaborative environments.